

MTH102 – Probability and Statistics
(2nd Mid-semester examination)

6th of March, 2026

Duration: 1 hour.

You **MUST** give **complete explanations** for all your answers.

Name: _____ Id: _____

1. A biased coin, which lands on head with probability 0.9, is tossed 10 times. Assume that the tosses are independent events. Let X denote the random variable that assigns the total number of heads.
 - (a) (1 point) What is $E[X]$?
 - (b) (1 point) What is $\text{Var}(X)$?
2. Let X be a random variable that represents the outcome of rolling a die. Let Y be a random variable that represents the outcome of rolling another die. Let $p(x, y)$ denote the joint probability mass function. Assume that X and Y are independent.
 - (a) (1 point) Compute $p(x, y)$ for each x, y .
 - (b) (1 point) Compute the marginal probability $p_Y(4)$.
 - (c) (1 point) Compute the conditional probability $p_{X|Y}(2 | 4)$ (i.e. $P_X = 2 | Y = 4$).
 - (d) (1 point) Let $Z = X + Y$. Compute $P\{Z = 4\}$.
 - (e) (1 point) Compute $E[Z]$.
 - (f) (1 point) Compute $\text{Var}[Z]$.
3. Recall that $\text{Cov}(X, Y) = E[(X - \mu_x)(Y - \mu_Y)]$. Show the following:
 - (a) (1 point) $\text{Cov}(X, X) = \text{Var}(X)$
 - (b) (1 point) $\text{Cov}(aX, Y) = a \text{Cov}(X, Y)$
 - (c) (2 points) $\text{Cov}(X_1 + X_2, Y_1 + Y_2) = \text{Cov}(X_1, Y_1) + \text{Cov}(X_1, Y_2) + \text{Cov}(X_2, Y_1) + \text{Cov}(X_2, Y_2)$
 - (d) (2 points) If X and Y are independent then $\text{Cov}(X, Y) = 0$
 - (e) (2 points) Assume that X and Y are identical **but not independent**, show that $\text{Cov}(X + Y, X - Y) = 0$.
4. (4 points) A biased coin lands on a head with probability 0.7. You toss it repeatedly. Use the law of large numbers to find out the minimum number of tosses required to be 99% certain that it lands on a head between 69% to 71% of the time?